



Bachelor of Technology Final year Project Evaluation

ROBUST DRONE IDENTIFICATION VIA RF SPECTROGRAM ANALYSIS USING ROBUSTSPECNET

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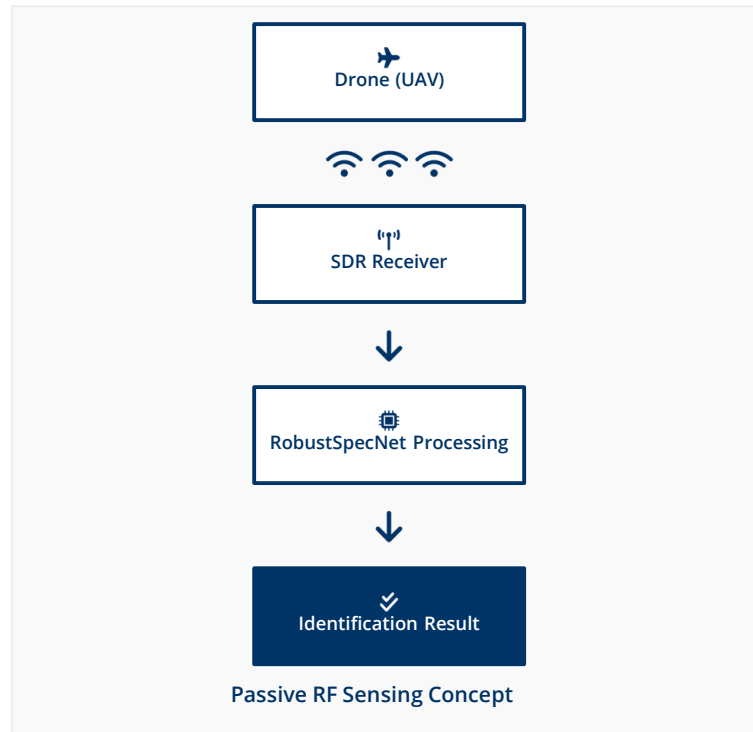
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INTRODUCTION

- Unmanned Aerial Vehicles (UAVs) are increasingly used in commercial, industrial, and public service applications [1], but their rapid growth raises critical security concerns regarding unauthorized airspace intrusion, privacy violations, and infrastructure threats [2].
- Traditional detection methods have significant limitations in detecting small, low-altitude drones [2].
- “” We propose RobustSpecNet, a deep learning framework using high-resolution RF spectrograms to identify DJI drone models, achieving 99.94% classification accuracy while maintaining low computational complexity [3], [5].

RobustSpecNet achieves 99.94% classification accuracy while maintaining low computational complexity.



PROBLEM STATEMENT

Environmental Challenges

- Massive challenges in noisy urban environments where RF signals are weak and distorted.
- Multipath fading and SDR hardware noise cause severe performance degradation.
- Signal quality is highly affected by distance variations and environmental interference.

Technical Limitations

- Existing models miss crucial fine-grained timing behavior of similar drone brands.
- Standard CNNs and Autoencoders struggle with subtle intra-model differences.
- Advanced transformer models demand excessive computational power, preventing real-time deployment.

OBJECTIVES OF THE PROJECT



System Development

Develop a lightweight AI-based system (RobustSpecNet) to accurately identify 17 different drone types using unique RF fingerprints.



Gain Invariance

Achieve distance-independent learning through precise signal normalization, ensuring robust performance across varying ranges.



Signal Preprocessing

Preprocess raw IQ RF signals via DC offset removal and Butterworth filtering to eliminate hardware interference.



Temporal Feature Extraction

Implement a custom Gated Unit (GUnit) specifically designed to capture fine-grained temporal patterns and pulse rhythms.

Raw Signal Collection



Clean & Normalize



Feature Transformation



Multi-Class Identification

LITERATURE REVIEW: LIMITATIONS

Early CNN & Autoencoders

- ✗ Rely on basic time-frequency features, missing complex signal nuances.
- ✗ Struggle to capture fine-grained differences between similar drone signals.
- ✗ Autoencoder methods are largely limited to binary classification (drone vs. non-drone).

Advanced Hybrid Models

- ✗ Transformer-based models require massive computational power and datasets.
- ✗ High architectural complexity makes real-time deployment difficult.
- ✗ Performance drops significantly when signals are highly similar (e.g., DJI variants).

Key Gap: Existing models fail to balance high classification accuracy with the low latency required for practical, real-time urban SDR environments.

LITERATURE REVIEW: CHALLENGES

Signal Complexity

- ✓ Difficulty in capturing fine-grained temporal patterns in RF signals [9], [11].
- ✓ Existing models focus heavily on signal shape rather than timing behavior [3], [9].
- ✓ Subtle intra-model differences are extremely hard to detect for similar drone types [3], [16].

Environmental Factors

- ✓ Urban SDR environments cause significant performance degradation [2], [5].
- ✓ Multipath fading and hardware noise severely affect identification accuracy [15], [12].
- ✓ Distance variations lead to signal strength fluctuations, impacting gain-sensitive models.

PROPOSED METHODOLOGY: PREPROCESSING

1 DC Offset Removal

Elimination of constant electrical offsets from raw IQ data to isolate true drone pulses.

2 Butterworth Filter

4th-order filter to suppress out-of-band noise.

50kHz - 1.1MHz

3 RMS Normalization

Equalization of signal power for distance-independent (gain-invariant) learning.



Signal Preprocessing Pipeline

METHODOLOGY: FEATURE TRANSFORMATION

STFT Generation

Conversion of 1D time-series IQ data into 2D high-resolution spectral images (Spectrograms) to capture rapid frequency hopping.

Hann Window

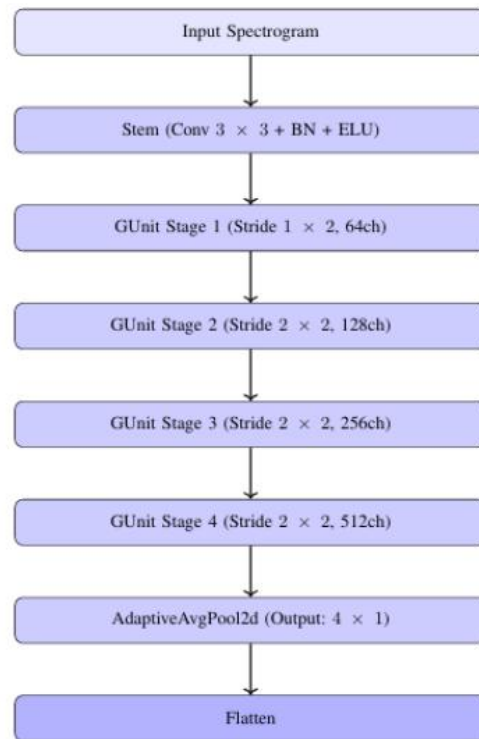
75% Overlap

Log-dB Scaling

Conversion of power levels to Decibels (dB) to highlight weak signal textures and patterns often lost in standard linear scales.

RF Fingerprinting

Creation of unique visual signatures for each drone model, enabling the deep learning model to perform spatial feature extraction.



SYSTEM ARCHITECTURE: ROBUSTSPECNET

Stem Layer

GUnit (Gated Unit) Innovation Implemented (1, 3) asymmetric kernels to prioritize the timing of signal bursts rather than just spatial shapes [9].

GUnit Stages (1-4)

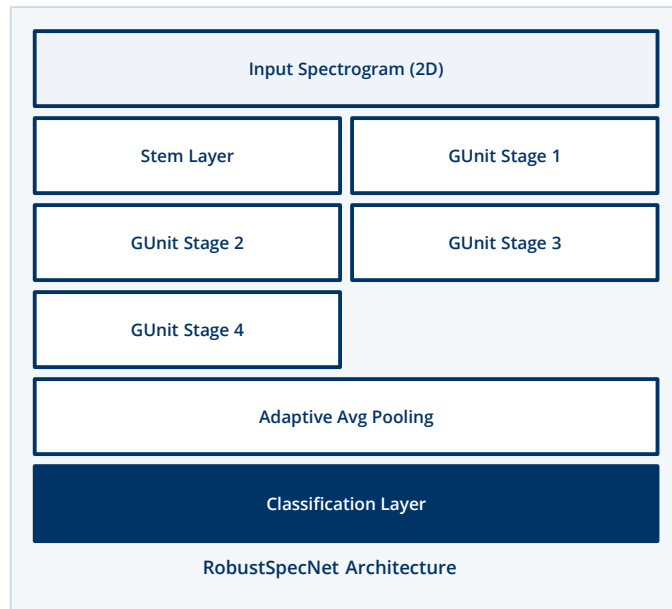
Core network component using asymmetric kernels to prioritize timing over spatial shape.

Frequency-Aware Pooling

Frequency-Aware Pooling Used AdaptiveAvgPool2d((4, 1)) to collapse time but keep 4 distinct frequency bands, preserving the drone's "spectral signature" [3], [5].

GUnit (Gated Unit) Innovation:

Implemented (1, 3) asymmetric kernels to capture fine-grained temporal





DATASET & TRAINING STRATEGY

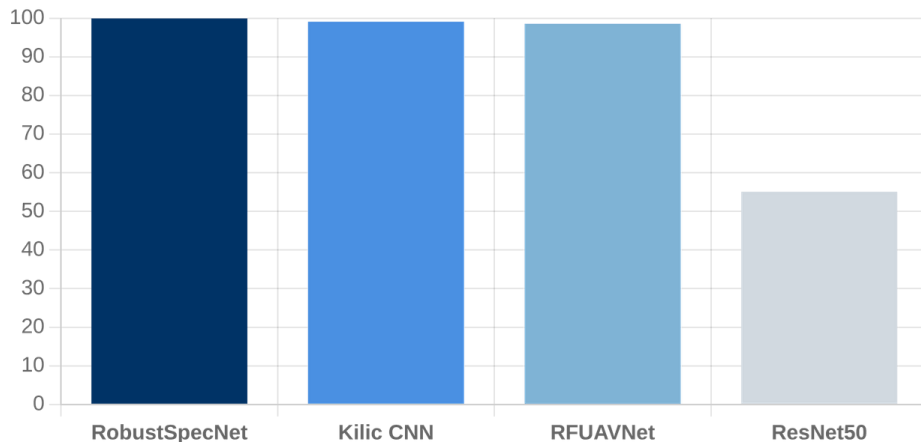
Dataset Composition

Metric	Value
Total Spectrograms	10,024
Drone Classes	17 (DJI & 3rd Party)
Training Samples (80%)	8,019
Testing Samples (20%)	2,005
Data Augmentation	None (Raw Data Only)

Training Parameters

- ✓ **Optimizer:** Adam optimizer with cross-entropy loss function.
- ✓ **Epochs:** Efficient training over just 20 epochs.
- ✓ **Overfitting Control:** Randomly disabled 50% of neurons (Dropout) during training.
- ✓ **Robustness:** Forced the model to learn drone features rather than background noise.

RESULTS AND ANALYSIS: PERFORMANCE



99.93%

PRECISION

99.95%

RECALL

99.94%

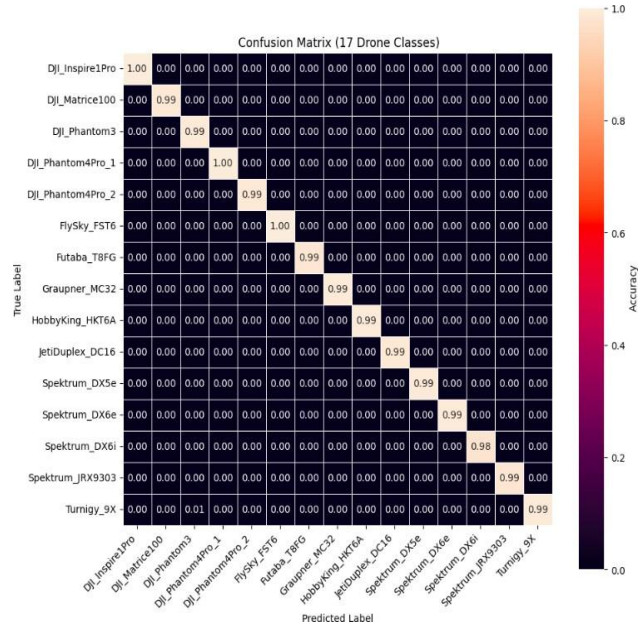
F1-SCORE

TABLE II
PERFORMANCE COMPARISON WITH BASELINE MODELS

Model	Input Type	Accuracy (%)	Key Optimization
RobustSpecNet	2D Spectrogram	99.94	GUnit + Adaptive Pooling
RFUAVNet [3]	1D Raw IQ	98.53	Grouped 1D Convolutions
Kilic <i>et al.</i> CNN [24]	2D Spectrogram	99.10	Combined Dense Layers
VGG11 BN [25]	2D Spectrogram	94.24	Deep Convolutional Baseline
ResNet50 [26]	2D Spectrogram	55.05	Deep Residual (High Parameters)

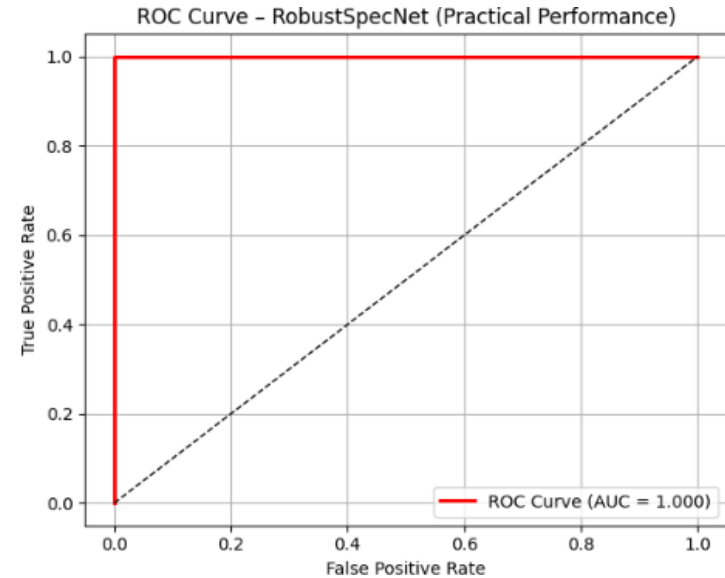
RESULTS AND ANALYSIS:

Confusion Matrix (17 Classes)



Analysis confirms **negligible misclassification** among 17 highly similar drone and controller types, demonstrating superior discriminative capability.

ROC Curve Analysis



Practical performance validated by an Area Under the Curve (AUC) approaching **perfection (1.000)**, indicating highly reliable detection.

CONCLUSION & FUTURE SCOPE

✓ Key Achievements

- 🏆 Developed **RobustSpecNet**, a lightweight framework achieving **99.94% accuracy** for passive RF drone identification.
- 🔧 Custom **Gated Unit (GUnit)** successfully captured critical time-frequency patterns, proving resilient against hardware distortions.
- 🛡️ Validated performance across 17 classes, outperforming state-of-the-art baselines in noisy urban SDR environments.

🚀 Future Directions

- 👥 Expand capabilities to achieve simultaneous identification of **multiple drone swarms** in crowded spectrums.
- 📱 Optimize the architecture for **low-power edge-device** deployment and hybrid sensing.
- 🔍 Investigate adversarial robustness and cross-platform signal generalization for broader drone model support.

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THANK YOU

Questions & Discussion



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